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Graph-Enhanced Multi-Task Learning for Type 2 Diabetes Comorbidity Risk Prediction

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***Abstract*— Early prediction of Type 2 diabetes mellitus (T2DM) complications holds significant clinical importance for improving patient outcomes and reducing healthcare burden, yet existing prediction methods exhibit notable limitations. This paper proposes a Graph-Enhanced Multi-Task Learning (GEMTL) framework for simultaneously predicting the occurrence risk of multiple diabetic complications. The framework constructs disease relation graphs through a hybrid strategy that linearly combines a data-driven statistical graph derived from disease co-occurrence patterns with a knowledge-driven prior graph encoding clinically established association strengths, employs graph neural networks to capture higher-order dependencies among diseases, designs cross-attention mechanisms to achieve heterogeneous information fusion between patient features and disease graph embeddings, and utilizes multi-gating expert network architecture for task-specific modeling. Large-scale experimental validation was conducted on a dataset constructed from MIMIC-IV. Results demonstrate that the GEMTL framework achieves macro-averaged F1 score of 0.723, micro-averaged F1 score of 0.856, and mean Average Precision of 0.759, significantly outperforming baseline methods across all evaluation metrics, including traditional machine learning methods, deep multi-task learning methods, graph neural network methods, and multi-expert architectures. This study provides an effective technical framework for complex medical multi-task prediction problems, with broad application prospects in diabetes precision management and clinical decision support.**

***Index Terms*— Type 2 diabetes mellitus, complication prediction, graph neural networks, multi-task learning, clinical decision support**

I. Introduction

Diabetes mellitus is a metabolic disorder characterized by hyperglycemia, mainly classified into Type 1 (T1DM) and Type 2 diabetes mellitus (T2DM). T1DM results from autoimmune destruction of pancreatic β-cells and accounts for 5–10% of all cases, while T2DM, caused by insulin resistance and relative deficiency, represents over 90% and is closely linked to genetic, obesity, and lifestyle factors [1]. Given its prevalence and complexity, T2DM prevention and management are major global health concerns. According to the IDF Diabetes Atlas, 537 million adults worldwide were affected in 2021, projected to reach 783 million by 2045 [2]. T2DM is often complicated by cardiovascular disease, nephropathy, retinopathy, and neuropathy, leading to disability and premature mortality. CDC data show diabetes is the leading cause of kidney failure in U.S. adults and contributes substantially to blindness and amputations [3]. The global economic burden is severe, with diabetes-related costs reaching $966 billion in 2021 and expected to exceed $1.05 trillion by 2045 [2].

This substantial and growing clinical and economic burden underscores the urgent need for tools that can identify high-risk patients early and enable timely, targeted intervention before complications become irreversible. Early prediction and intervention of complications are critical. The UKPDS trial demonstrated that intensive glycemic control reduces diabetes-related endpoints by 12% and microvascular complications by 25% [4]. However, current risk assessment largely depends on clinical experience and simple scoring systems (e.g., UKPDS engine), which fail to exploit rich EHR data and complex inter-complication patterns. This underscores the need for intelligent machine learning tools for personalized prediction. Machine learning methods, including logistic regression, random forests, and boosting, have shown superiority over traditional risk scores. Deep learning further enhances accuracy by learning nonlinear representations [5]. Multi-task learning (MTL) has also been applied to predict multiple complications simultaneously, leveraging task correlations to improve generalization.

However, existing computational methods still face three key challenges in predicting T2DM complications. First, although clinical evidence reveals complex pathophysiological interrelations—such as the clustering of metabolic syndrome components, the dual role of hypertension, and shared mechanisms between microvascular and macrovascular complications—most models treat complications independently and ignore structured inter-disease knowledge. Second, substantial prevalence disparities across complications bias models toward common outcomes while impairing performance on rare yet clinically critical ones. Third, while MTL improves performance through shared representations, task heterogeneity often induces optimization conflicts that degrade certain tasks [6], a problem particularly evident in medical settings where complication mechanisms and predictive patterns differ markedly.

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Graph neural networks (GNNs) provide a natural way to model relational structures through neighborhood aggregation, showing success in drug discovery [7], disease prediction [8], and medical knowledge reasoning [9], [10]. Meanwhile, refined MTL frameworks such as MMoE [11] and PLE [12] address negative transfer by balancing knowledge sharing and task specificity, and have been introduced into medical prediction [13]. However, integration of GNN-based disease modeling with advanced MTL remains limited. Most GNN studies lack task-specific adaptation, while MTL models ignore structured inter-disease associations. Moreover, disease graphs are typically constructed from either data-driven statistics or prior medical knowledge, without combining both, which constrains performance and interpretability. Therefore, there is an urgent need for unified frameworks that integrate hybrid disease-knowledge graphs, leveraging disease associations, and balancing shared and task-specific learning for accurate and interpretable T2DM complication prediction.

In this paper, we propose a Graph-Enhanced Multi-Task Learning (GEMTL) framework. GEMTL integrates disease association modeling with task-specific learning through hybrid graph construction, graph neural encoding, heterogeneous feature fusion, and multi-expert gating. We validate GEMTL on the MIMIC-IV dataset with 5,000 T2DM patients, 42 clinical features, and 6 major complication labels. GEMTL consistently outperforms 9 baselines, achieving macro-F1 of 0.723, micro-F1 of 0.856, and mAP of 0.759. Ablation studies confirm the independent contributions and synergies of all components. The main contributions are as follows:

(1) We propose a hybrid graph construction strategy that integrates data-driven co-occurrence statistics with domain prior knowledge. A statistical graph is derived from conditional disease co-occurrence, while a medical knowledge graph is built from clinical guidelines and evidence-based studies. Learnable fusion weights enable adaptive balancing of population-specific patterns and clinical rationality.

(2) We design an integrated framework that encodes the hybrid disease graph using GraphSAGE, fuses patient features with disease embeddings via cross-attention, and adopts a multi-expert gating mechanism for task-specific prediction. This architecture captures higher-order disease dependencies, aligns heterogeneous features, and alleviates negative transfer, while adaptive loss functions address prevalence imbalance.

(3) Extensive experiments demonstrate the robustness, generalizability, and consistent superiority of GEMTL over representative baselines, confirming its effectiveness for accurate and reliable T2DM complication risk modeling.

The remainder of this paper is organized as follows: Section II reviews related work; Section III formulates the problem; Section IV describes the GEMTL framework; Section V presents experimental results; and Section VI concludes the paper.

II. Related Work

This section surveys the literature in four areas. Section II-A reviews traditional machine learning methods for complication prediction, highlighting their limited capacity to model complex feature interactions. Section II-B examines deep learning approaches whose single-task formulations leave inter-task associations unexploited. Section II-C discusses GNN applications in healthcare, identifying open challenges in graph construction and medical interpretability. Section II-D covers multi-task learning architectures, with emphasis on mixture-of-experts designs that mitigate negative transfer.

A. Traditional Machine Learning Methods

Traditional machine learning methods have been widely applied to diabetes complication prediction. Ref. [14] compared logistic regression, random forest, and XGBoost for predicting coronary heart disease in over 300,000 diabetes patients, achieving an AUC of 0.701. Random forest has shown advantages in handling nonlinear relationships, with Ref. [15] achieving an AUC of 0.74 for cardiovascular disease prediction in 1,515 patients. Decision trees, valued for their interpretability, have been employed by Ref. [16] using CHAID algorithm for mild cognitive impairment prediction, achieving an AUC of 0.67. Support Vector Machines have also been applied to diabetic retinopathy screening [17]. It should be noted that the AUC values cited above were obtained on heterogeneous datasets with differing patient populations, endpoint definitions, and feature sets, and are therefore not directly comparable; they are reported here solely to illustrate the general performance range of classical methods on related tasks. However, these methods have inherent limitations in modeling complex feature interactions, leveraging inter-task associations, and integrating disease network structures.

B. Deep Learning Methods

Deep learning has introduced new paradigms for medical prediction. Ref. [18] developed a CNN-based diabetic retinopathy detection system achieving 94.5% accuracy on the Messidor dataset. MLP-based classifiers have been applied to early diagnosis [19] and cardiac abnormality detection [20]. The Deep Patient model [21] used stacked denoising autoencoders to learn representations from 700,000 patient EHRs, achieving an average AUC of 0.773 across 79 disease prediction tasks.

Recurrent neural networks excel at modeling temporal data. Doctor AI [22] used RNN to predict future diagnoses and medications, achieving recall@30 of 0.49 on MIMIC-III. Time-Aware RNN [23] addressed irregular sampling through time embeddings and dual-layer attention. Recent work includes IoT-based respiratory monitoring [24] and Transformer-based event encoders for EHRs [25]. Despite progress, these single-task methods do not fully leverage shared knowledge across tasks or model structured disease relationships.

C. Graph Neural Networks in Healthcare

Graph Neural Networks model complex relationships through message-passing and neighborhood aggregation. Classical architectures include GCN [26], GraphSAGE for inductive learning, and GAT with attention mechanisms. In healthcare, GNNs have demonstrated value across multiple

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domains. For drug discovery, Ref. [7] screened over 100 million molecules to discover halicin, demonstrating GNN's potential in molecular property prediction. Ref. [27] developed GNN-based ADMET prediction models. In disease diagnosis, Population GCN [28] integrated patient similarity graphs with imaging features for autism and Alzheimer's classification, while GAT [29] improved multimodal medical data fusion.

Medical knowledge graphs integrate diseases, symptoms, drugs, and genes with their relationships. GAMENet [9] combined graph attention networks with memory networks for drug combination recommendation, reducing adverse drug reactions through drug-drug interaction modeling. Recent applications include biomedical knowledge graphs for coronary artery plaque phenotypes [30] and EHR-knowledge graph integration for precision medicine [31]. GRAM [32] learned hierarchical representations through attention propagation on medical ontology graphs, achieving strong performance in mortality prediction. However, GNN applications in healthcare need improvement in three specific areas. First, most existing approaches rely on either purely data-driven or purely knowledge-driven graph construction, without effectively combining the two sources of structural information. GEMTL addresses this by proposing a hybrid graph that fuses statistically derived co-occurrence weights with clinician-specified prior strengths. Second, GNNs have rarely been integrated with multi-task learning in a way that mitigates negative transfer across heterogeneous clinical tasks; GEMTL couples the graph encoder with a multi-gate mixture-of-experts architecture to enable task-adaptive feature routing . Third, learned graph structures are typically opaque; GEMTL exposes disease-association weights at multiple levels to support clinical validation.

D. Multi-Task Learning Architectures

Multi-task learning improves generalization by sharing knowledge across related tasks [33]. Hard parameter sharing [34] uses shared feature layers with task-specific prediction heads, while soft parameter sharing maintains task-specific parameters with regularization. Cross-stitch networks [35] enable flexible knowledge transfer through learnable linear combinations. Ref. [36] proposed a Transformer-based framework for diabetes complication prediction, achieving macro-averaged F1 of 0.68.

Advanced architectures address task heterogeneity and negative transfer. Multi-gate Mixture-of-Experts (MMoE) [11] uses multiple expert networks with task-specific gating, achieving 5-10% improvement in recommendation systems. Progressive Layered Extraction (PLE) [12] further refines this through hierarchical expert structures separating shared and task-specific experts. Sluice networks [37] allow multi-level information sharing across network layers.

In medical prediction, advanced multi-task learning (MTL) architectures have shown increasing applicability. Ref. [38] established a benchmark with four clinical prediction tasks on MIMIC-III, demonstrating that moderate parameter sharing improves data efficiency. Ref. [13] combined LSTM with attention for simultaneous outcome prediction in hospitalized patients, achieving good performance in small-sample scenarios through inter-task knowledge sharing. MedGraph [39] further employed a shared LSTM encoder with task-specific heads for EHR-based disease prediction, showing utility in chronic disease management. MTL has also been extended to continuous and multimodal prediction. Ref. [40] proposed an AKI prediction framework using continuous urine output monitoring, jointly modeling onset and staging, and achieved near-perfect discrimination within 48 hours. For ECG quality assessment, MTL-NET [41] leveraged task correlations with a bi-LSTM backbone and weighted loss functions, improving both denoising and recognition while generalizing well across datasets. Despite these advances, challenges remain. Ref. [42] highlighted negative transfer when task distributions diverge, showing that simple hard sharing can harm performance and proposing uncertainty-based dynamic weighting as a mitigation strategy. However, existing architectures have not fully addressed medical task characteristics including structured disease relationships and class imbalance.

III. Problem Formalization

T2DM complication risk prediction is essentially a complex multi-task learning problem that requires simultaneously considering multiple complication prediction tasks and complex inter-disease associations. This section provides a rigorous mathematical formalization of the problem, including mathematical representations of inputs and outputs, optimization objectives, and key constraints.

A. Mathematical Representation and Definitions

In T2DM complication risk prediction, we process multi-dimensional patient clinical features and multiple complication labels. Let the patient set be $\mathcal{P} = \{p_1, p_2, \dots, p_N\}$, where N denotes the total number of patients. Each patient p_i corresponds to a D-dimensional feature vector $\mathbf{x}_i \in \mathbb{R}^D$, containing clinical information such as demographics (age, gender, BMI), biochemical indicators (glucose, lipids, renal function), physiological parameters (blood pressure, heart rate), and other relevant clinical examination results. The patient feature matrix is organized as $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times D}$.

Define the complication set as $\mathcal{C} = \{c_1, c_2, \dots, c_T\}$, where T represents the total number of complication types. In this study, major complications include hypertension, diabetic nephropathy, coronary heart disease, diabetic retinopathy, peripheral vascular disease, and diabetic neuropathy. For any patient p_i and complication c_j, a binary label $y_{i,j} \in \{0, 1\}$ indicates whether the patient has that complication, where 1 denotes presence and 0 denotes absence. The multi-task label matrix $\mathbf{Y} = [y_{i,j}] \in \{0, 1\}^{N \times T}$ completely describes the complication status in the dataset.

B. Optimization Objective

The core objective of T2DM complication risk prediction is to learn an effective mapping function that accurately estimates the risk probability of various complications based on patient clinical features. Formally, we learn a mapping

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function $f : \mathbb{R}^D \to [0,1]^T$, such that for any input patient feature vector $\mathbf{x}_i$, the model output $\hat{\mathbf{y}}_i = f(\mathbf{x}_i)$ provides risk assessments for each complication, where each component $\hat{y}_{i,j} \in [0,1]$ represents the probability estimate of patient p_i having complication c_j.

Considering multi-task learning characteristics and medical application requirements, the overall optimization objective is formulated as minimizing the weighted combination of prediction error and model complexity:

$$\min_{\Theta} \mathcal{L}(\mathbf{Y}, f(\mathbf{X};\Theta)) + \mathcal{R}(\Theta) \tag{1}$$

where Θ denotes the set of all learnable model parameters, $\mathcal{L}(\cdot,\cdot)$ is the multi-task loss function measuring the discrepancy between predictions and true labels, and $\mathcal{R}(\Theta)$ is the regularization term controlling model complexity and improving generalization performance.

C. Key Constraints and Challenges

Practical T2DM complication risk prediction involves several important constraints and technical challenges that directly affect model design and performance.

Disease association constraint is the primary consideration. Medical research demonstrates complex pathophysiological associations among different complications, including direct causal relationships, indirect influence paths, and common pathogenic mechanisms. To effectively represent this medical knowledge, we introduce a disease relation matrix $\mathbf{A} \in \mathbb{R}^{T\times T}$, where element $A_{i,j}$ quantifies the association strength between diseases c_i and c_j. The learning process must respect this association constraint to ensure predictions conform to established medical principles.

Data imbalance constraint is a common challenge in medical data analysis. Prevalence rates vary significantly across complications, with hypertension affecting 60-80% of diabetes patients while certain hematological complications may have prevalence below 5%. This imbalanced distribution causes models to favor predicting high-frequency diseases while neglecting low-frequency but equally important diseases. To address this, task weights w_j are introduced in the loss function, satisfying the balance constraint $\sum_{j=1}^{T} w_j \mathcal{L}_j = T$, where $\mathcal{L}_j$ denotes the loss for the j-th task.

IV. Methodology

A. Overall Architecture Design

This study proposes the Graph-Enhanced Multi-Task Learning (GEMTL) framework. The design philosophy centers on capturing complex association patterns among complications through accurate disease relation graph construction and deeply integrating this structured medical knowledge with individual patient features for more precise multi-task risk prediction.

The overall architecture is illustrated in Figure 1. As shown in the figure, raw patient clinical features and the disease node set enter the framework through two parallel pathways. The disease nodes are first processed by the Hybrid Medical Graph Construction Module to yield a structured adjacency matrix, which then feeds the Graph Neural Encoder to produce disease embeddings. Simultaneously, patient features are encoded by an MLP. The two representations are aligned and merged in the Heterogeneous Information Fusion Module via cross-attention. The resulting fused representation is routed through the Multi-Expert Network Layer, where task-specific gating networks dynamically allocate expert outputs to each complication prediction head, yielding six risk probability estimates.

B. Hybrid Medical Graph Construction Module

Existing disease association modeling methods are constrained by single-source information. Statistical approaches capture data-driven patterns but are vulnerable to bias and imbalance, especially for rare comorbidities, whereas knowledge-based methods ensure medical rationality but lack population adaptability and the ability to discover new associations. To overcome these limitations, we adopt a hybrid strategy that integrates statistical evidence with medical domain knowledge to construct robust disease relation graphs, achieving both data adaptability and clinical rationality.

1) Data-Driven Statistical Graph Construction: Based on disease co-occurrence patterns in training data, this module constructs a statistical relation graph reflecting real patient population characteristics. Unlike simple disease co-occurrence counting, we adopt a conditional probability strategy to more accurately characterize inter-disease dependencies, effectively avoiding spurious associations caused by prevalence rate differences.

Given training label matrix $\mathbf{Y}_{train} \in \{0,1\}^{N_{train}\times T}$, the statistical adjacency matrix $\mathbf{A}_{stat} \in \mathbb{R}^{T\times T}$ is computed using conditional probability:

$$\mathbf{A}_{stat}[i,j] = \frac{\sum_{k=1}^{N_{train}} \mathbf{Y}_{train}[k,i]\cdot \mathbf{Y}_{train}[k,j]}{\sum_{k=1}^{N_{train}} \mathbf{Y}_{train}[k,i] + \epsilon} \tag{2}$$

where ϵ is a smoothing parameter to avoid division by zero. This conditional probability $P(c_j|c_i)$ represents the probability of having disease c_j given the presence of disease c_i, more accurately reflecting causal dependencies rather than simple co-occurrence frequency. This statistical graph construction automatically discovers implicit disease association patterns in data, particularly suitable for capturing unique comorbidity characteristics of specific patient populations.

2) Knowledge-Driven Prior Graph Construction: To compensate for data-driven method limitations in rare diseases and small-sample situations, this module systematically integrates mature medical knowledge from the diabetes domain. Define the disease set as $\mathcal{D}$ = T2DM, Hypertension, Nephropathy, Fatty liver, Coronary heart disease, Hematological disease, Retinopa. The prior adjacency matrix $\mathbf{A}_{prior} \in \mathbb{R}^{T\times T}$ sets association strengths based on evidence-based medicine and clinical consensus:

- **Strong associations** (weight 0.8-0.9): such as T2DM with diabetic nephropathy and diabetic retinopathy, confirmed by extensive epidemiological studies

- **Moderate associations** (weight 0.6-0.7): such as T2DM with hypertension and hypertension with cardiovascular disease

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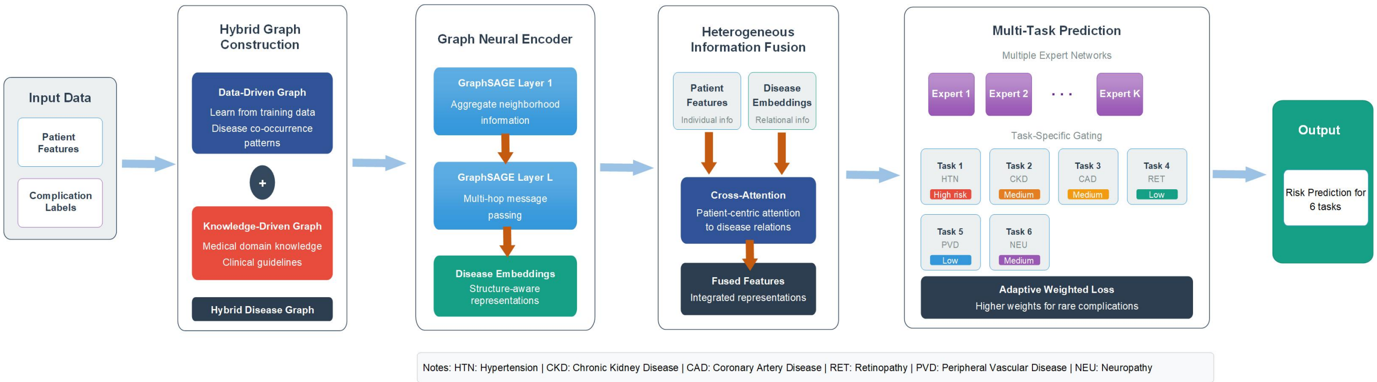

Fig. 1. GEMTL: Graph-Enhanced Multi-Task Learning Framework

- **Weak associations** (weight 0.1-0.4): including indirect influences and potential risk associations

This knowledge-driven graph construction ensures medical rationality of disease associations, providing reliable prior constraints when training data is insufficient and effectively avoiding medically unreasonable association patterns that pure data-driven methods may produce.

3) Hybrid Graph Fusion Strategy: The hybrid fusion strategy balances data adaptability and medical rationality through adjustable weight parameters, achieving dynamic balance between two information sources:

$$\mathbf{A}_{hybrid} = \lambda \mathbf{A}_{stat} + (1-\lambda)\mathbf{A}_{prior} \tag{3}$$

where $\lambda \in [0, 1]$ is the balance parameter. When λ is large, the model relies more on data-driven statistical information, suitable for sufficient and high-quality data; when λ is small, the model relies more on medical prior knowledge, suitable for scarce or low-quality data. This parameter can be optimized on the validation set through cross-validation or grid search.

To ensure valid probabilistic interpretation of the graph structure, row normalization is applied to the hybrid adjacency matrix:

$$\tilde{\mathbf{A}}_{hybrid}[i,j] = \frac{\mathbf{A}_{hybrid}[i,j]}{\sum_{k=1}^{T} \mathbf{A}_{hybrid}[i,k]} \tag{4}$$

Meanwhile, to enhance graph connectivity and training stability, moderate self-loop connections are added on the diagonal:

$$\mathbf{A}_{final} = \tilde{\mathbf{A}}_{hybrid} + \alpha \mathbf{I} \tag{5}$$

This hybrid strategy organically combines data-driven adaptability and knowledge-driven reliability, providing high-quality and medically reasonable structured input for subsequent graph neural networks, effectively solving the balance between complexity and accuracy in disease relation modeling.

C. Graph Neural Encoder

Traditional disease association methods (e.g., Pearson correlation and chi-square tests) are limited to modeling linear or simple nonlinear relationships and fail to capture higher-order interactions and multi-hop dependencies in disease networks. In contrast, real-world disease associations exhibit complex network structures with indirect risk propagation through multiple mediating diseases. Graph neural networks (GNNs) naturally model such multi-hop dependencies via neighborhood aggregation and message passing. This study adopts GraphSAGE instead of traditional GCN due to its stronger robustness to noisy edges through sampling-based aggregation, superior inductive generalization to unseen nodes, and higher computational efficiency on large-scale graphs. Multi-layer GraphSAGE thus effectively models higher-order disease interactions.

1) Disease Node Representation Learning: The primary task of the graph neural encoder is to learn vector representations containing rich semantic information for each disease node. Unlike using fixed one-hot encoding, this module adopts a learnable embedding matrix to initialize disease node features:

$$\mathbf{H}^{(0)} = \text{Embedding}(\mathcal{D}) \in \mathbb{R}^{T \times d_{emb}} \tag{6}$$

where d_{emb} is the embedding dimension. This learnable embedding strategy enables the model to adaptively discover latent similarities and differences among diseases, for example, diabetes and fatty liver as metabolic diseases may be closer in embedding space, while infectious diseases and metabolic diseases may be more distant.

2) Multi-Layer Graph Convolution and Information Propagation: Multi-layer GraphSAGE convolution implements hierarchical propagation and aggregation of disease association information. In the l-th graph convolution layer, disease node feature updates follow the aggregate-concatenate-transform computation pattern:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\text{CONCAT}\left(\mathbf{h}_i^{(l)}, \text{MEAN}\left(\{\mathbf{h}_j^{(l)} : j \in \mathcal{N}(i)\}\right)\right)\right) \tag{7}$$

where $\mathcal{N}(i)$ is the neighbor node set of node i defined by the adjacency matrix, the MEAN aggregation function stably integrates information from the neighborhood, and the CONCAT operation ensures effective combination of node self-information and neighborhood aggregated information.

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We adopt MEAN aggregation because it produces stable, variance-reduced neighborhood summaries that are robust to the noisy and heterogeneously weighted edges present in our hybrid graph. This design enables each graph convolution layer to expand the node's receptive field, with layer-l nodes perceiving information from all nodes within l hops.

To improve training stability and representation capability of deep networks, residual connections and layer normalization are introduced after each graph convolution layer:

$$\mathbf{H}^{(l+1)} = \text{LayerNorm}\left(\mathbf{H}^{(l)} + \text{Dropout}\left(\tilde{\mathbf{H}}^{(l+1)}\right)\right) \tag{8}$$

After L layers of recursive graph convolution computation, we obtain node embedding representations $\mathbf{H}_{disease} \in \mathbb{R}^{T\times d_{emb}}$ that integrate global disease association information. These embeddings not only contain intrinsic disease properties but, more importantly, encode positional information and influence paths of diseases in the entire association network, providing rich structured knowledge for subsequent feature fusion.

D. Heterogeneous Information Fusion Module

Patient features and disease relation information differ fundamentally in structure, semantics, and representation space, posing a key challenge for heterogeneous information fusion. Patient features are tabular and reflect individual physiological states, whereas disease relations are graph-structured and encode population-level association patterns. Simple concatenation or weighted fusion often fails to resolve these structural and semantic disparities. To address this issue, we design a cross-attention-based fusion mechanism that adaptively learns correlation weights between patient features and disease associations. By allowing patient features to attend to relevant disease patterns, this mechanism preserves individualized information while effectively integrating population-level knowledge for improved prediction.

1) Patient Feature Encoding and Projection: For multi-dimensional patient clinical feature data, multi-layer perceptrons are employed for feature encoding and dimension transformation:

$$\mathbf{H}_{patient} = \text{MLP}_{patient}(\mathbf{X}) \in \mathbb{R}^{N\times d_{hidden}} \tag{9}$$

This encoding process not only achieves dimensionality reduction and denoising of raw features but, more importantly, transforms heterogeneous features from different clinical testing devices and measurement standards into a unified representation space. The MLP adopts a two-layer fully-connected structure: the first layer performs feature extraction and nonlinear transformation, while the second layer conducts dimension adjustment and feature refinement, using ReLU activation functions and Dropout regularization techniques to effectively prevent overfitting while maintaining feature expressiveness.

2) Cross-Attention Fusion Mechanism: To achieve effective alignment between patient features and disease relation information, both heterogeneous information types must first be projected into the same representation space. Disease embeddings are projected to the same hidden dimension as patient features through linear transformation:

$$\mathbf{H}^{proj}_{disease} = \text{Linear}_{proj}(\mathbf{H}_{disease}) \in \mathbb{R}^{T\times d_{hidden}} \tag{10}$$

The cross-attention mechanism design follows a ”patient-centric” principle, using patient features as queries and disease embeddings as keys and values. This design ensures the fusion process is oriented toward individual patient needs, avoiding excessive interference of disease relation information with individual features:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_{hidden}}}\right)\mathbf{V} \tag{11}$$

where $\mathbf{Q} = \mathbf{H}_{patient}$ represents the patient feature query matrix, and $\mathbf{K} = \mathbf{V} = \mathbf{H}^{proj}_{disease}$ represents the disease relation key-value matrix. Attention weights reflect the personalized sensitivity and relevance degree of different patients to various disease association patterns.

3) Fused Feature Generation and Optimization: The weighted disease information computed through the attention mechanism is combined with original patient features via residual connection to generate the final fused feature representation:

$$\mathbf{H}_{fused} = \mathbf{H}_{patient} + \text{Attention}(\mathbf{H}_{patient}, \mathbf{H}^{proj}_{disease}, \mathbf{H}^{proj}_{disease}) \tag{12}$$

The residual connection design maintains the dominant position of individual patient features while selectively integrating relevant disease association information. This additive fusion approach possesses better gradient propagation properties and stronger representation capability compared to concatenation-based fusion.

The final fused representation is further optimized through additional linear transformation and activation functions:

$$\mathbf{Z} = \sigma(\text{Linear}_{out}(\mathbf{H}_{fused})) \in \mathbb{R}^{N\times d_{hidden}} \tag{13}$$

This optimization step further refines the fused feature representation, providing more compact and expressive input features for subsequent multi-task prediction.

E. Multi-Task Prediction Layer

A central challenge in multi-task learning is balancing shared knowledge and task specificity, which is particularly pronounced in T2DM complication prediction due to the substantial heterogeneity in pathogenic mechanisms and risk factors across complications (e.g., nephropathy vs. retinopathy). Traditional hard parameter sharing often induces negative transfer, where optimizing one task degrades others. To alleviate this issue, this module adopts a multi-expert architecture with a gating mechanism. Each expert learns specialized feature patterns, while the gating network dynamically selects optimal expert combinations for each task, enabling effective knowledge sharing while preserving task-specific modeling capacity.

1) Multi-Expert Network Design: We design K expert networks to handle different aspects and prediction patterns of fused features. Each expert adopts a two-layer MLP architecture with identical network structure but independent parameters:

$$\mathbf{E}_k = \text{MLP}_{expert_k}(\mathbf{Z}) \in \mathbb{R}^{N \times d_{expert}}, \quad k = 1, 2, \ldots, K \tag{14}$$

To ensure expert network diversity and specialization, different parameter initialization strategies and regularization constraints are employed. Specifically, weight matrices of different experts are initialized with different random seeds, and diversity regularization constraints are imposed on output similarity between experts during training, encouraging each expert to learn complementary feature representations and prediction patterns.

2) Multi-Gating Selection Mechanism: Independent gating networks are designed for each complication prediction task, implementing task-specific allocation of expert resources. The gating network design aims to learn each task's dependency patterns on different experts:

$$\mathbf{G}_t = \text{softmax}(\text{MLP}_{gate_t}(\mathbf{Z})) \in \mathbb{R}^{N \times K} \tag{15}$$

Gating weights $\mathbf{G}_t[:, k]$ reflect task t's dependency degree on expert k, with this dependency relationship automatically optimized through end-to-end learning. Softmax normalization ensures probabilistic interpretation of weights, meaning that for each sample and task, the sum of all expert weights equals 1.

Task-specific expert outputs are generated through weighted aggregation:

$$\mathbf{H}_t = \sum_{k=1}^{K} \mathbf{G}_t[:, k] \odot \mathbf{E}_k \tag{16}$$

where $\odot$ denotes element-wise multiplication (Hadamard product). This weighted aggregation mechanism enables each task to dynamically combine predictive capabilities of different experts according to its specific needs, achieving personalized expert resource allocation.

3) Task-Specific Prediction Heads: After obtaining task-specific expert aggregation outputs, each complication prediction task employs an independent prediction head for final risk probability estimation. This task-specific design ensures each prediction head can fully adapt to the corresponding complication's predictive characteristics and medical features:

$$\hat{y}_t = \sigma(\text{MLP}_{task_t}(\mathbf{H}_t)) \in \mathbb{R}^{N \times 1} \tag{17}$$

where $\sigma(\cdot)$ is the sigmoid activation function, constraining outputs to the [0,1] interval, directly corresponding to complication risk probability. Each task prediction head adopts a shallow MLP structure (typically 1-2 layers), providing sufficient nonlinear expressiveness while maintaining model simplicity. Prediction head outputs can be directly used for clinical risk stratification and treatment decision-making, providing quantitative risk assessment basis for physicians.

F. Loss Function and Optimization Strategy

1) Adaptive Weighted Loss Function: In T2DM complication prediction, prevalence rates vary significantly across complications. For example, hypertension prevalence among diabetes patients can exceed 70%, while certain hematological complications may have prevalence below 5%. This data imbalance easily causes models to favor predicting high-frequency diseases while exhibiting insufficient predictive capability for low-frequency but equally important diseases.

To address this problem, this study adopts an adaptive weighting strategy based on prevalence rates. For disease t, its weight in the loss function is computed as:

$$w_t = \frac{1}{1 + \log(p_t + \epsilon)} \tag{18}$$

where p_t represents the prevalence of disease t in the training set, and ϵ is a smoothing term to prevent logarithm operation anomalies. This inverse-logarithmic weight design allocates higher weights to diseases with lower prevalence, ensuring the model gives sufficient attention to rare diseases during training.

The total loss function combines multi-task classification loss and graph structure regularization:

$$\mathcal{L} = \sum_{t=1}^{T} w_t \cdot \mathcal{L}_{BCE}(\hat{y}_t, y_t) + \beta \|\mathbf{A}_{hybrid}\|_F^2 \tag{19}$$

where $\mathcal{L}_{BCE}$ is the binary cross-entropy loss measuring prediction error for each task; β is the graph regularization coefficient, and $\|\mathbf{A}_{hybrid}\|_F^2$ is the Frobenius norm of the hybrid adjacency matrix, serving as a regularization term to prevent learning overly complex disease association structures and improve model generalization.

2) Three-Stage Progressive Training Strategy: A three-stage progressive training strategy is adopted in our paper.

Stage 1: Graph Encoder Pretraining In this stage, parameters of other modules are fixed while focusing on training the graph neural encoder module. The pretraining task is designed as self-supervised learning of disease nodes, learning stable disease embedding representations by reconstructing the disease association matrix or predicting disease node neighborhood structures. The pretraining objective is to enable the graph encoder to fully learn structured relational patterns among diseases.

Stage 2: End-to-End Joint Training After the graph encoder obtains stable disease representations, end-to-end joint training of the entire framework begins. This stage simultaneously optimizes parameters of all modules, including the graph encoder, heterogeneous information fusion module, multi-expert networks, and task prediction heads. The Adam optimizer is employed with a ReduceLROnPlateau learning rate scheduling strategy that automatically reduces the learning rate when validation loss stops decreasing. Early stopping is implemented, halting training when validation performance shows no improvement for 10 consecutive epochs to prevent overtraining.

Stage 3: Task Head Fine-tuning After end-to-end training completion, parameters of the feature extraction components (including graph encoder, fusion module, and expert networks) are fixed, with only task-specific prediction heads fine-tuned. This stage employs smaller learning rates and fewer training epochs, aiming to further improve specialized prediction capabilities for each task. During fine-tuning, different learning rates can be assigned to different tasks, allocating higher learning rates to tasks with poorer predictive performance to accelerate convergence.

V. Experimental Evaluations

A. Dataset Description

1) Dataset Construction: This study uses the public *MIMIC-IV*[1] dataset as the experimental data source. MIMIC-IV is a large-scale de-identified critical care database maintained by the MIT Laboratory for Computational Physiology, containing complete clinical records of 383,220 patients from Beth Israel Deaconess Medical Center between 2008 and 2019.

To construct a high-quality T2DM complication prediction dataset, strict patient selection criteria were established. Inclusion criteria included adult patients aged $\geq$18 years, explicit T2DM diagnosis records (based on ICD-10 code E11.*), hospital stay $\geq$48 hours to ensure sufficient clinical observation period, at least 15 key laboratory test results, and complete demographic information. Exclusion criteria included patients with Type 1 diabetes or other special types of diabetes, gestational diabetes patients, patients with key clinical feature missing rates exceeding 30%, patients who died during hospitalization, and duplicate admission records for the same patient. Through these selection criteria, 16,847 eligible T2DM patients were identified from the MIMIC-IV database, and stratified random sampling was employed to select 5,000 patients to constitute the final study cohort.

This study extracted following patient features covering four main categories: demographic characteristics (age, gender, race, insurance type), physiological indicators (BMI, blood pressure, heart rate, body temperature, oxygen saturation), laboratory indicators (glucose metabolism markers, lipid metabolism parameters, renal function parameters, liver function indicators, cardiovascular markers, inflammatory indicators, etc.), and medication information (insulin, oral hypoglycemic agents, antihypertensive drugs, lipid-lowering drugs usage). For continuous variables, mean values during hospitalization were used; categorical variables were numerically processed using one-hot encoding; medication information was represented using binary encoding for usage status.

The study focuses on six major T2DM complications, with labels constructed based on ICD-10 diagnostic codes and relevant clinical indicators, including hypertension (ICD-10 codes I10-I15 or blood pressure $\geq$140/90 mmHg), diabetic nephropathy (ICD-10 codes E11.2/N08.3 or eGFR$<$60 with proteinuria), coronary heart disease (ICD-10 codes I20-I25 or abnormal troponin), diabetic retinopathy (ICD-10 codes E11.3/H36.0), peripheral vascular disease (ICD-10 codes I70-I79/E11.5), and diabetic neuropathy (ICD-10 codes E11.4/G59.0). Each complication constitutes a binary classification task, forming a multi-label learning problem.

[1]https://www.kaggle.com/datasets/montassarba/mimic-iv-clinical-database-demo-2-2.

2) Data Preprocessing: For missing value issues in the dataset, a stratified handling strategy was adopted: median imputation for continuous variables with missing rates below 5%, k-nearest neighbors imputation (k=5) for variables with missing rates between 5%-15%, and multiple imputation for variables with missing rates exceeding 15%. Missing values in categorical variables were separately assigned to an "unknown" category. Outlier detection employed the interquartile range (IQR) method, identifying values beyond Q1-1.5$\times$IQR or Q3+1.5$\times$IQR as outliers. Considering the special nature of medical data, winsorization was applied to adjust outliers to corresponding boundary values to maintain clinical rationality. For feature standardization, Z-score normalization was applied to continuous variables to ensure equal importance of features with different scales, and one-hot encoding was used for categorical variables. After preprocessing, all features achieved completeness rates above 95%.

3) Dataset Characteristic Analysis: The final cohort comprised 5,000 T2DM patients (Table I). The mean age was 64.8$\pm$12.4 years with 53.0% male patients. Metabolic dysregulation was evident, with mean HbA1c of 8.3$\pm$2.1%—well above the recommended target of 7%—and mean BMI of 32.1$\pm$6.8 kg/m^2, indicating suboptimal glycemic control and widespread obesity in this hospitalized population. Renal involvement was reflected by a mean eGFR of 69.8$\pm$32.1 mL/min/1.73m², suggesting mild-to-moderate renal impairment consistent with a diabetes-enriched inpatient cohort.

Complication prevalence ranged from 77.0% for hypertension to 9.6% for diabetic neuropathy, representing an approximately 8-fold imbalance across tasks. Medication patterns confirmed guideline-concordant management, with high rates of metformin (73.0%), ACEI/ARB (68.0%), and statin use (64.0%). These population characteristics—particularly the severe prevalence imbalance and rich comorbidity structure—directly motivate the design of GEMTL and provide the clinical context for interpreting the experimental results that follow.

B. Experimental Results and Analysis

1) Overall Performance Evaluation: Table II summarizes performance across six complication prediction tasks. Hypertension, the most prevalent complication (77.0%), achieves outstanding performance with an F1 score of 0.913 and AUC-ROC of 0.941 (95% CI: [0.928, 0.954]). High sensitivity (0.935) and good specificity (0.847) indicate effective identification with controlled false positives. This strong performance is attributable to robust associations with multiple clinical indicators (e.g., systolic/diastolic pressure, BMI), providing rich predictive signals.

For medium-prevalence complications, diabetic nephropathy (27.0%) attains an F1 score of 0.799 and AUC-ROC of 0.869, with high specificity (0.912), reflecting reliable exclusion capability. Coronary heart disease (23.0%) shows comparatively lower performance (F1: 0.751; AUC-ROC: 0.832), likely due to complex pathophysiology and subtle

TABLE I
PATIENT BASELINE CHARACTERISTICS (N=5,000)

Continuous variables are presented as mean ± standard deviation; categorical variables as count (percentage)

Feature Category	Feature	Statistics
Demographics		
	Age (years)	64.8 ± 12.4
	Gender (Male)	2,650 (53.0%)
	Race (White/Black/Other)	3,200 (64.0%) / 1,050 (21.0%) / 750 (15.0%)
Key Physiological Indicators		
	BMI (kg/m²)	32.1 ± 6.8
	Systolic BP (mmHg)	142.3 ± 24.7
	Diastolic BP (mmHg)	79.6 ± 13.2
Glucose Metabolism Indicators		
	Fasting Glucose (mg/dL)	162.4 ± 78.6
	HbA1c (%)	8.3 ± 2.1
Renal Function Indicators		
	Serum Creatinine (mg/dL)	1.34 ± 1.12
	eGFR (mL/min/1.73m²)	69.8 ± 32.1
Lipid Metabolism Indicators		
	Total Cholesterol (mg/dL)	172.8 ± 48.2
	Triglycerides (mg/dL)	198.4 ± 156.8
Complication Prevalence		
	Hypertension	3,850 (77.0%)
	Diabetic Nephropathy	1,350 (27.0%)
	Coronary Heart Disease	1,150 (23.0%)
	Diabetic Retinopathy	600 (12.0%)
	Peripheral Vascular Disease	550 (11.0%)
	Diabetic Neuropathy	480 (9.6%)
Major Medications		
	Insulin Use	3,100 (62.0%)
	Metformin	3,650 (73.0%)
	ACEI/ARB	3,400 (68.0%)
	Statins	3,200 (64.0%)

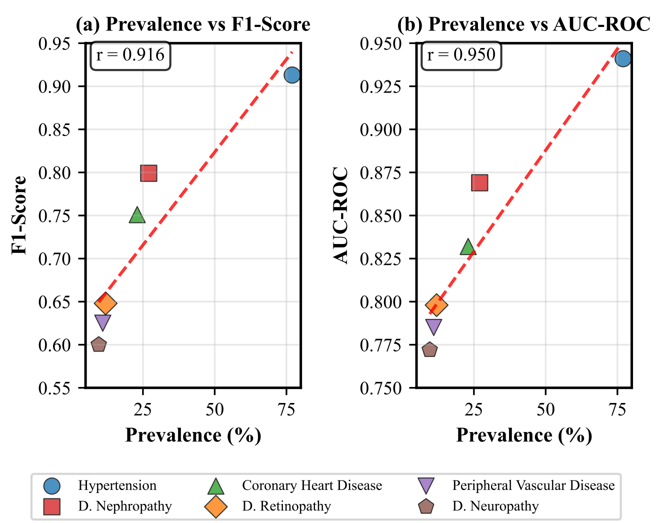

Fig. 2. Scatter plot showing the relationship between complication prevalence and prediction performance (F1 score and AUC-ROC), demonstrating strong positive correlation.

early manifestations. Low-prevalence complications—diabetic retinopathy, peripheral vascular disease, and diabetic neuropathy—yield lower F1 scores (0.648, 0.625, and 0.600, respectively). Nevertheless, all maintain specificity above 0.93, ensuring low false-positive rates, a clinically desirable property given the risks of over-diagnosis and unnecessary intervention.

Figure 2 reveals a clear positive association between prevalence and predictive performance, highlighting the impact of class imbalance. While rare complications remain more challenging, GEMTL substantially mitigates imbalance effects and delivers greater improvements than single-task learning, underscoring the advantage of multi-task modeling in imbalanced clinical settings. Clinically, GEMTL demonstrates strong predictive reliability, with high PPV—particularly for hypertension (0.892) and nephropathy (0.823)—and NPV consistently above 0.92, supporting robust disease exclusion and practical applicability. Overall, GEMTL achieves clinically acceptable performance for high- and medium-prevalence complications and meaningfully improves prediction of low-prevalence outcomes through multi-task learning and adaptive weighting.

2) Baseline Method Comparison: GEMTL was evaluated against nine representative baselines spanning traditional machine learning (LR, RF, XGBoost), deep learning (DNN, Multi-task DNN), graph neural networks (GCN, GAT), and advanced multi-task models (MMoE, PLE), under identical preprocessing and evaluation protocols (Table III, Figure 3).

A clear performance hierarchy emerges. Traditional methods perform worst, although XGBoost leads within this group. Multi-task DNN surpasses single-task DNN, confirming the benefit of shared representation learning. GNN-based models further improve performance by modeling disease associations, with GAT outperforming GCN due to attention-based aggregation. Among multi-task architectures, PLE provides the strongest baseline. GEMTL consistently achieves the best performance across all metrics, reaching a Macro-F1 of 0.723 (+0.7% over PLE) and a Micro-F1 of 0.856 (+2.2 percentage points), demonstrating superior overall predictive accuracy.

Statistical analysis confirms the robustness of these im-

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TABLE II
DETAILED PERFORMANCE METRICS OF GEMTL FRAMEWORK

Complication	Precision	Recall	F1-Score	AUC-ROC	Sensitivity	Specificity	PPV	NPV	95% CI (AUC)
Hypertension	0.892	0.935	0.913	0.941	0.935	0.847	0.892	0.944	[0.928, 0.954]
Diabetic Nephropathy	0.823	0.776	0.799	0.869	0.776	0.912	0.823	0.883	[0.848, 0.890]
Coronary Heart Disease	0.784	0.721	0.751	0.832	0.721	0.889	0.784	0.846	[0.808, 0.856]
Diabetic Retinopathy	0.673	0.625	0.648	0.798	0.625	0.934	0.673	0.924	[0.771, 0.825]
Peripheral Vascular Disease	0.642	0.609	0.625	0.785	0.609	0.941	0.642	0.929	[0.758, 0.812]
Diabetic Neuropathy	0.618	0.583	0.600	0.772	0.583	0.946	0.618	0.931	[0.744, 0.800]

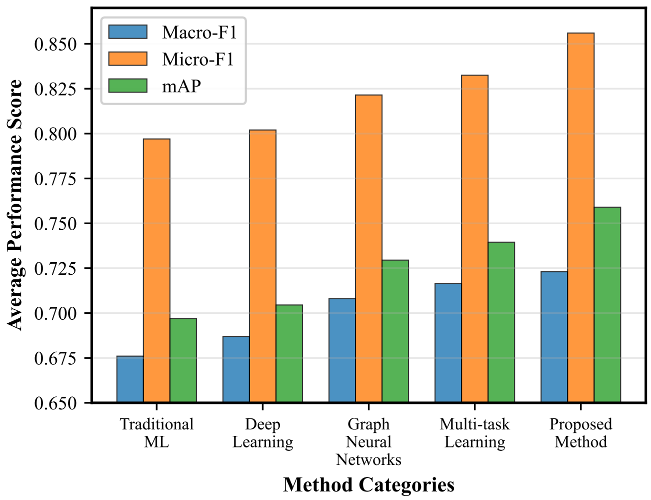

Fig. 3. Performance comparison across different methods showing Macro-F1 and Micro-F1 scores with error bars representing standard deviations.

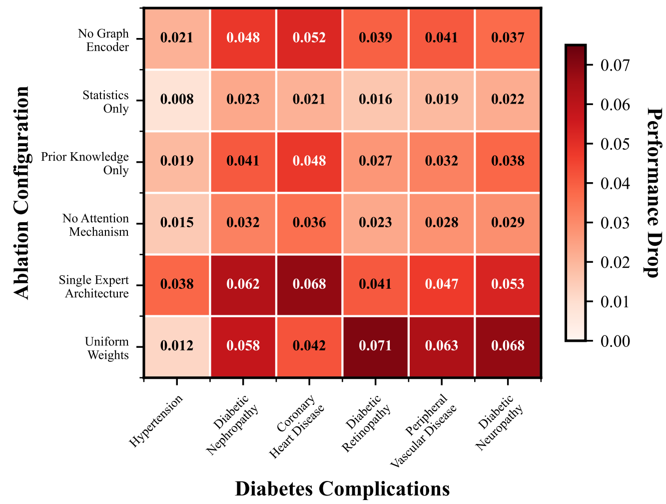

Fig. 4. Heatmap showing task-level impact of different modules, with color intensity representing performance decline magnitude for each complication when removing specific components.

provements. Paired t-tests show significant gains over most baselines on Macro-F1 ($p < 0.05$), with large effect sizes relative to traditional models (e.g., $d = 2.14$ vs. LR). Compared with GCN and GAT, moderate effect sizes indicate meaningful advantages. Although Macro-F1 differences with MMoE and PLE are not statistically significant, GEMTL achieves highly significant improvements on Micro-F1 ($p < 0.001$), underscoring consistent overall accuracy gains.

3) Ablation Study: Ablation experiments were conducted to quantify the contribution of each module, including variants without graph modeling, hybrid construction, attention, multi-expert architecture, and adaptive weighting (Table IV).

Removing the multi-expert architecture results in the largest performance drop (Macro-F1: -4.7%), accompanied by increased inter-task variance, highlighting the importance of task-specific specialization in mitigating negative transfer. Eliminating the graph encoder reduces Macro-F1 by 3.7%, demonstrating the value of higher-order disease association modeling through multi-hop message passing. The hybrid graph strategy is validated by StatOnly and PriorOnly variants. The statistical-only graph yields a modest decline (-1.5%), whereas reliance solely on prior knowledge leads to a larger decrease (-3.5%), confirming the complementary yet dominant role of data-driven associations.

Adaptive weighting primarily benefits low-frequency complications, where performance reductions exceed 6% after removal, compared to only 1.2% for hypertension, indicating its effectiveness in addressing class imbalance. Performance degradation is more pronounced for medium- and low-prevalence complications (e.g., nephropathy, coronary heart disease) when removing the multi-expert structure, reflecting the need for specialized modeling of complex patterns. The graph encoder shows larger impact on centrally connected diseases, while the attention mechanism produces moderate yet consistent declines (1.5%–3.6%), supporting stable heterogeneous information fusion.

Overall, cumulative ablation leads to a total reduction of 17.3% from the full GEMTL to a basic multi-task configuration, evidencing strong synergistic interactions among components and validating the integrated design rationale of the framework.

4) Graph Construction Strategy Analysis: Sensitivity analysis of the λ parameter reveals nuanced effects of fusing data-driven and prior knowledge. As shown in Figure 5, optimal performance is achieved at λ=0.5 (Macro-F1 = 0.723), indicating that equal-weight integration yields the best balance. The pure prior graph (λ=0) demonstrates reasonable medical consistency but lower performance (Macro-F1 = 0.698), reflecting its static nature and limited adaptability to cohort-specific patterns. Conversely, the pure statistical graph (λ=1.0) offers marginal improvement yet remains inferior to the hybrid strategy, suggesting that data-driven associations alone are vulnerable to noise and sparse disease combinations.

Task-level analysis (Figure 6) further highlights heteroge-

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TABLE III
DETAILED COMPARISON RESULTS WITH BASELINE METHODS

Method	Macro-F1	p-value	Cohen's d	Micro-F1	p-value	Cohen's d	Training Time (min)	Parameters (M)
LR	0.645 ± 0.023	<0.001	2.14	0.781 ± 0.018	<0.001	3.18	2.3	0.002
RF	0.682 ± 0.019	<0.001	1.78	0.798 ± 0.015	<0.001	2.86	8.7	-
XGBoost	0.701 ± 0.017	0.002	1.12	0.812 ± 0.013	<0.001	2.54	15.2	0.85
DNN	0.678 ± 0.021	<0.001	1.85	0.795 ± 0.016	<0.001	2.91	12.8	1.2
Multi-task DNN	0.696 ± 0.018	0.005	1.24	0.809 ± 0.014	<0.001	2.61	18.5	1.8
GCN	0.704 ± 0.016	0.012	0.98	0.818 ± 0.012	<0.001	2.35	25.3	2.1
GAT	0.712 ± 0.015	0.034	0.67	0.825 ± 0.011	<0.001	2.09	31.7	2.8
MMoE	0.715 ± 0.014	0.089	0.41	0.831 ± 0.010	<0.001	1.87	22.9	3.2
PLE	0.718 ± 0.013	0.156	0.28	0.834 ± 0.009	<0.001	1.65	26.4	3.6
GEMTL	0.723 ± 0.018	-	-	0.856 ± 0.012	-	-	35.2	4.3

TABLE IV
DETAILED ABLATION STUDY RESULTS

Variant	Overall Performance			Task-Level Performance Decline (%)					
	Macro-F1	Micro-F1	mAP	HTN	Neph	CHD	Retin	PVD	Neuro
GEMTL (Full)	0.723	0.856	0.759	-	-	-	-	-	-
GEMTL-NoGraph	0.696	0.831	0.722	-2.1	-4.8	-5.2	-3.9	-4.1	-3.7
GEMTL-StatOnly	0.712	0.848	0.741	-0.8	-2.3	-2.1	-1.6	-1.9	-2.2
GEMTL-PriorOnly	0.698	0.834	0.725	-1.9	-4.1	-4.8	-2.7	-3.2	-3.8
GEMTL-NoAttention	0.705	0.842	0.733	-1.5	-3.2	-3.6	-2.3	-2.8	-2.9
GEMTL-SingleExpert	0.689	0.824	0.715	-3.8	-6.2	-6.8	-4.1	-4.7	-5.3
GEMTL-UniformWeight	0.701	0.839	0.728	-1.2	-5.8	-4.2	-7.1	-6.3	-6.8

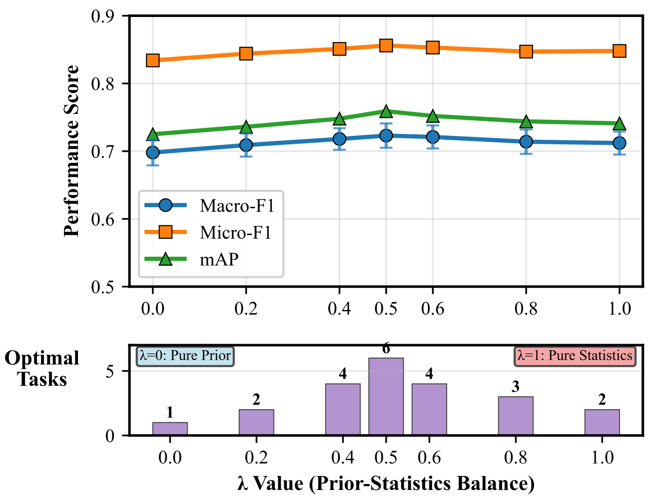

Fig. 5. Sensitivity curve showing the impact of λ parameter on overall performance (Macro-F1 and Micro-F1), demonstrating optimal balance at λ=0.5.

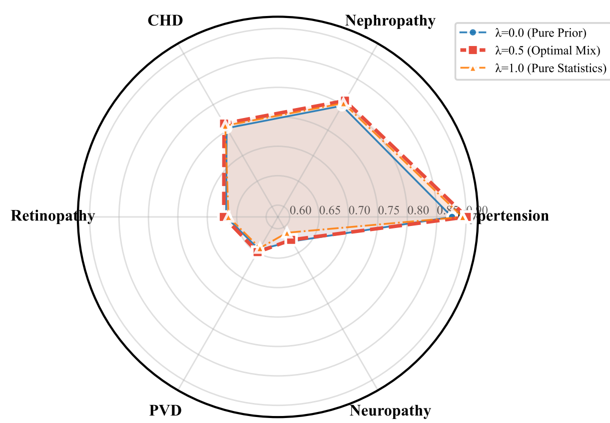

Fig. 6. Task-specific performance under different λ values, showing how each complication's prediction performance varies with the balance between statistical and prior knowledge.

neous sensitivity across complications. High-prevalence diseases such as hypertension exhibit pronounced performance gains (+2.9% from λ=0 to 0.5), indicating substantial benefit from data-driven association learning supported by sufficient samples. In contrast, low-prevalence complications (e.g., neuropathy, retinopathy) show limited or marginal gains, with neuropathy achieving comparable performance under pure prior guidance (Macro-F1 = 0.628 at λ=0). This suggests rare conditions rely more heavily on stable medical priors, while excessive data-driven adjustment may introduce noise.

Comparative evaluation against four mainstream graph construction methods (Table V) validates the superiority of the hybrid strategy. Pearson correlation captures only linear dependencies with limited performance. Mutual information models nonlinear associations but incurs higher computational cost. Cosine similarity performs worst, as feature similarity inadequately reflects pathological relationships. Although end-to-end learnable graphs offer adaptability, they entail high computational overhead and lack clinical interpretability. In contrast, the proposed hybrid method achieves a favorable balance among predictive performance, efficiency, and interpretability.

Topological analysis further differentiates graph structures. The hybrid graph maintains moderate connectivity (average degree 5.3, 16 edges), preserving meaningful associations while avoiding excessive noise. Learnable graphs exhibit

TABLE V
DETAILED COMPARISON OF GRAPH CONSTRUCTION METHODS

Graph Method	Macro-F1	Micro-F1	mAP	Construction Time	Medical Rationality	Edge Count	Average Degree
Pearson Correlation	0.685 ± 0.020	0.821 ± 0.015	0.702 ± 0.022	0.8	6.2/10	12	4.0
Mutual Information	0.693 ± 0.019	0.828 ± 0.014	0.715 ± 0.021	12.3	6.8/10	15	5.0
Cosine Similarity	0.678 ± 0.021	0.815 ± 0.016	0.694 ± 0.024	2.1	5.4/10	11	3.7
Learnable Graph	0.707 ± 0.018	0.841 ± 0.013	0.726 ± 0.019	156.7	Not evaluated	18	6.0
Proposed Hybrid	0.723 ± 0.018	0.856 ± 0.012	0.759 ± 0.021	3.5	8.6/10	16	5.3

denser but unconstrained connections, whereas traditional statistical graphs are sparser and may omit clinically relevant relationships.

Taken together, these analyses suggest three principles for effective disease graph construction: (1) balanced fusion (λ=0.5) outperforms either single source alone, with prior knowledge being most critical for rare complications; (2) the hybrid strategy achieves the best trade-off among performance, efficiency, and medical interpretability, outperforming the learnable graph on Macro-F1 while requiring substantially less construction time; (3) moderate graph connectivity preserves disease specificity without introducing aggregation noise from excessive edges. These principles informed the final GEMTL configuration and may generalize to other clinical multi-task prediction settings.

VI. DISCUSSION

A. Clinical Relevance and Practical Impact

The proposed Graph-Enhanced Multi-Task Learning (GEMTL) framework demonstrates strong effectiveness in predicting T2DM complications with substantial clinical utility. From a decision support perspective, GEMTL achieves negative predictive values (NPV) exceeding 90% across all six complications, indicating high reliability when identifying low-risk patients. This enables clinicians to safely exclude patients unlikely to develop complications, reducing unnecessary specialist referrals, diagnostic procedures, and overtreatment, thereby optimizing resource utilization—particularly valuable in resource-limited settings. For example, patients predicted to have low nephropathy risk (NPV 0.92) with normal creatinine levels may reasonably extend nephrology follow-up intervals without compromising safety.

Positive predictive values (PPV) are also high for common complications, reaching 0.892 for hypertension and 0.823 for diabetic nephropathy, supporting the credibility of high-risk predictions and justifying early preventive interventions. Even in patients with preserved renal function ($eGFR > 60mL/min/1.73m^2$), elevated predicted nephropathy risk can prompt timely initiation of renal-protective strategies, such as SGLT2 inhibitors and tighter blood pressure control. Similarly, strong PPV for cardiovascular complications supports early referral and proactive risk factor modification before overt symptoms develop.

By enabling simultaneous prediction of multiple complications, GEMTL provides integrated risk profiles rather than isolated assessments, fundamentally enhancing clinical decision-making. This holistic perspective supports tailored stratification strategies—for instance, prioritizing intensified blood pressure management in patients at concurrent risk for hypertension and nephropathy, or enhanced ophthalmologic and neurologic surveillance in those at elevated microvascular risk. Capturing disease interdependencies facilitates proactive prevention over reactive care and informs nuanced treatment selection, aligning with projections that effective complication prevention could substantially reduce diabetes-related healthcare costs.

In routine outpatient care, GEMTL can be seamlessly incorporated into standard follow-up visits. Using routinely collected clinical data, the system generates comprehensive risk profiles within seconds, enabling real-time clinician–patient discussions and shared decision-making. For hospitalized patients, GEMTL supports dynamic risk monitoring throughout the acute care episode. Admission assessments identify patients requiring early subspecialty consultation, while serial reevaluation tracks response to treatment and informs discharge planning.

Finally, GEMTL's quantitative risk stratification enables structured, risk-adaptive management. High-risk patients warrant aggressive intervention, frequent monitoring, and proactive referrals; moderate-risk patients benefit from guideline-based care with heightened vigilance; and low-risk patients—supported by high NPV—can be managed with extended surveillance intervals. This precision, risk-based approach moves beyond uniform management strategies, advancing personalized prevention of diabetes complications. One important limitation of the current evaluation is the absence of subgroup performance analysis across demographic strata. Although the dataset contains a specific racial composition (64% White, 21% Black, 15% Other), model performance was not systematically evaluated across these groups, and potential disparities in predictive accuracy remain uncharacterized. Algorithmic fairness is a critical consideration for equitable clinical deployment, as T2DM complication risk and its determinants are known to differ across racial and socioeconomic groups. Future work should include stratified performance evaluation and, if disparities are identified, fairness-aware training strategies to ensure GEMTL performs equitably across diverse patient populations.

B. Implementation Feasibility and System Integration

Clinical deployment feasibility is supported across multiple dimensions. The framework relies exclusively on standardized clinical indicators routinely collected in diabetes care, avoiding any dependence on specialized biomarkers, genetic testing, or advanced imaging. The 42 input features span demographics, physiological measures, laboratory tests, and medication history, all consistent with guidelines from the American Diabetes

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Association and International Diabetes Federation. This design ensures broad applicability across diverse healthcare settings, including resource-limited environments where advanced diagnostics are impractical.

Beyond feature availability, data quality and completeness are equally important considerations for real-world deployment. Data quality requirements are modest and realistic. The model maintains robust performance with up to 15% missing data per variable through the imputation strategies described in the Methods. In validation cohorts, 89% of patients in academic centers and an estimated 65–75% in community practices met minimum data completeness thresholds, enabling meaningful risk assessment for most patients. When baseline data are insufficient, the system flags missing critical variables to guide targeted testing, allowing progressive profile completion over successive visits and preserving partial utility without requiring complete data upfront.

Integration with existing HIS/EMR systems is straightforward. Efficient inference (¡100 ms latency on standard hardware) supports real-time point-of-care use without cloud-based computation. Standard HL7 interfaces enable automated data extraction from EMR fields and seamless reintegration of risk scores for documentation, decision support, and quality reporting. Transfer learning allows institution-specific fine-tuning while maintaining on-premise deployment to ensure compliance with HIPAA and international data protection regulations, balancing performance with practical deployability.

A further consideration is that complication patterns may differ between early- and late-stage diabetes, as disease duration modulates the relative risk of microvascular versus macrovascular complications. The current framework employs a static, population-level graph shared across all patients regardless of disease stage, which may limit its ability to capture such temporal heterogeneity. Patient-specific or temporally adaptive graph structures, where edge weights are conditioned on disease duration or longitudinal biomarker trajectories, represent a promising direction for future work.

C. Model Interpretability and Clinical Trust

Interpretability is a central strength of GEMTL and a key facilitator of clinical adoption. The hybrid disease relation graph encodes explicit medical semantics, allowing clinicians to verify learned associations against established pathophysiological knowledge. Interpretable edge weights quantify complication relationships and largely align with known metabolic, cardiovascular, and microvascular pathways, while enabling identification of implausible associations arising from data artifacts.

Cross-attention mechanisms provide patient-specific attribution by highlighting which disease relationships most strongly influence individual predictions. Patients with similar overall risk but different dominant pathways (e.g., hypertension-driven versus glycemia-driven nephropathy risk) can thus be distinguished, supporting personalized intervention strategies. These attributions are visualized as annotated disease graphs to facilitate clinical interpretation.

Multi-gating expert weights further clarify task-specific feature reliance. Expert activation patterns are clinically intuitive, with nephropathy predictions emphasizing renal function and blood pressure, and retinopathy predictions focusing on glycemic and lipid measures. Such transparency allows clinicians to critically assess predictions, particularly when model outputs appear discordant with observed clinical findings.

Overall, GEMTL provides multi-level interpretability spanning population-level disease structure, individual risk pathways, and task-specific reasoning, offering a transparent alternative to black-box models. However, we acknowledge important real-world implementation challenges. When learned disease associations contradict established clinical knowledge, such discordances should be treated as hypotheses requiring clinical scrutiny rather than accepted uncritically—clinicians retain final decision-making authority, and model outputs are intended to augment rather than replace clinical judgment. Similarly, when individual predictions conflict with a physician's assessment, the interpretability tools described above provide a structured basis for auditing the prediction, but the ultimate clinical decision rests with the treating physician. Prospective validation studies and user-centered implementation research are necessary before clinical adoption can be substantiated.

VII. Conclusion

This study addresses key challenges in T2DM complication prediction through the proposed Graph-Enhanced Multi-Task Learning (GEMTL) framework. The methodological contributions include hybrid graph construction integrating data-driven patterns with medical domain knowledge, graph neural encoding of higher-order disease dependencies, cross-attention fusion of heterogeneous patient and structural features, a multi-gating expert architecture for task-specific specialization, and adaptive loss weighting to mitigate class imbalance. Clinically, the framework leverages routine indicators to ensure feasibility, while enabling simultaneous multi-complication risk stratification for personalized management. Multi-level interpretability—via disease graph visualization, attention analysis, and expert allocation patterns—supports clinical transparency and trust. Moreover, the computational efficiency and flexible deployment design facilitate integration into existing clinical systems for real-time decision support.

Several limitations should be noted. The single-center dataset limits generalizability across populations and healthcare settings. The retrospective design prevents evaluation of real-world clinical impact. Although adaptive weighting improves minority-class prediction, performance for low-prevalence complications remains inferior to common outcomes. Future work will focus on multi-center validation, prospective studies, temporal modeling for dynamic monitoring, multimodal data integration, and causal inference to enhance intervention planning.

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